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Review

Autoimmune comorbidity in chronic spontaneous urticaria: A systematic review

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ABSTRACT

Background and objective: Numerous autoimmune diseases (AIDs) have been linked to chronic spontaneous urticaria (CSU). Here, we provide the first extensive and comprehensive evaluation of the prevalence of AIDs in patients with CSU and *vice versa*.
Methods: A Pubmed and Google Scholar search was performed to identify studies reporting the prevalence of various AIDs in CSU and *vice versa* published before April 2017.
Results: The prevalence of individual AIDs in CSU is increased (≥1% in most studies vs ≤1% in the general population). AIDs with relatively high prevalence in the general population are also quite common in CSU patients, whereas those with low prevalence remain a rare finding in CSU. The rates of comorbidity in most studies were ≥1% for insulin-dependent diabetes mellitus, rheumatoid arthritis (RA), psoriasis and celiac disease (CD), ≥2% for Graves' disease, ≥3% for vitiligo, and ≥5% for pernicious anemia and Hashimoto's thyroiditis. Organ-specific AIDs are more prevalent in CSU than systemic (multiorgan or non organ-specific) AIDs. >2% of CSU patients have autoimmune polyglandular syndromes encompassing autoimmune thyroid disease (ATD) and vitiligo or pernicious anemia. Antithyroid and antinuclear antibodies are the most prevalent AID-associated autoantibodies in CSU. >15% of CSU patients have a positive family history for AIDs. The prevalence of urticarial rash in AID patients is >1% in most studies. This rash is more prevalent in eosinophilic granulomatosis with polyangiitis, ATD, systemic lupus erythematosus, RA and CD.
Conclusions: CSU patients have an increased risk of AIDs, especially adult female patients and those with a positive family history and a genetic predisposition for AIDs, who should be screened for signs and symptoms of AIDs.
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Abbreviations: AID, autoimmune disease; ATD, autoimmune thyroid disease; ANA, antinuclear antibodies; Anti-dsDNA, antibodies against double-stranded DNA; Anti-Sm, anti-Smith antibodies; Anti-SSA, antibodies against the Ro/SSA antigens; Anti-SSB, antibodies against the La/SSB antigens; APS, autoimmune polyglandular (polyendocrine) syndrome; EGPA, eosinophilic granulomatosis with polyangiitis; CSU, chronic spontaneous urticaria; FH, family history; GD, Graves' disease; HT, Hashimoto's thyroiditis; IDDM, insulin-dependent diabetes mellitus (diabetes mellitus type 1); MAS, multiple autoimmune syndromes; RA, rheumatoid arthritis; RF, rheumatoid factor; SLE, systemic lupus erythematosus.

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1. Introduction

Chronic spontaneous urticaria (CSU) is a mast cell-driven disease characterized by the development of wheals, angioedema (AE), or both for >6 weeks [1]. The etiopathogenesis of CSU is thought to be associated with autoimmune mechanisms [2]. Numerous autoimmune diseases (AIDs) have been linked to CSU [3], most of which are complex chronic inflammatory disorders that involve a plethora of cells and mediators, affect many tissues and organs, and cause substantial morbidity and mortality, mostly in women aged 40 to 50 years [4,5].

AIDs can be classified as systemic (multiorgan or non organ-specific) and organ-specific [6,7]. Some systemic AIDs are primary, i.e. they occur alone, while secondary AIDs develop during the course of other AIDs, e.g. Sjögren's syndrome in patients with rheumatoid arthritis or systemic lupus erythemasosus. The etiopathogenesis of AIDs involves many factors that include, but are not limited to, predisposing genes, epigenetic alterations, hormonal effects, and environmental triggers, leading from premorbid B and T cell dysregulation to the development of clinically evident disease [8].

Recently, we systematically reviewed the association of CSU and autoimmune thyroiditis [9] and of CSU and systemic lupus erythematosus [10], and we found that CSU is strongly linked to both of these AIDs. Here, we provide the first extensive and comprehensive evaluation of the prevalence of AIDs in patients with CSU and *vice versa*.

2. Methods

A Pubmed search was performed to identify studies published before April 2017 with the keywords "urticaria" and "chronic urticaria" in combination with the terms that are listed in Table 1. No language filter was applied. We also examined the reference lists of the retrieved articles and we searched Google Scholar with the same keywords to identify additional studies. In total, 64 studies were included in this systematic review. Most of these studies looked at whether or not the patients included had AID at the time they were assessed (point prevalence). A few studies reported whether patients had a history of AID or not. In some studies, it was not specified if the patients still had the AID at the time they had their urticaria. Some studies evaluated the prevalence of more than one AID. The selection process of relevant studies is summarized in Table 1. In studies identified, we did a search for the prevalence of autoantibodies (antinuclear, rheumatoid factor, anti-dsDNA, anti-parietal cell, anti-transglutaminase, anti-smooth muscle, anti-mitochondrial, anti-SSA, anti-SSB, anti-Sm, anti-cardiolipin) and a family history of AIDs. The term "urticarial rash" was used for studies where CSU was not clearly differentiated from acute urticaria, chronic inducible urticaria and/or urticarial vasculitis.

3. Results

3.1. All investigated autoimmune diseases occur in >1% of CSU patients in most studies

In CSU patients, the prevalence of the 21 investigated individual AIDs ranged from 0% to 27.5% (Tables 2, 5). Almost all studies (95%) found each investigated AID in one or more CSU patient. AIDs with relatively high prevalence in the general population [11] proved to be also quite common in CSU patients, whereas those with low prevalence remained a rare finding in CSU group (Tables 2, 5). For example, the prevalence of comorbidity in most studies were ≥1% for insulin-dependent diabetes mellitus (IDDM), rheumatoid arthritis (RA), psoriasis and celiac disease, ≥2% for Graves' disease (GD), ≥3% for vitiligo, and ≥5% for pernicious anemia and Hashimoto's thyroiditis (HT) (Fig. 1).

3.2. Organ-specific AIDs are more prevalent in CSU than systemic AIDs

Prevalence of individual organ-specific AIDs and systemic AIDs in CSU ranged from 0 to 27.5% in 60 studies (including 24 studies on ATD previously analyzed [9]) and 0–4.7% in 40 studies, respectively (Tables 2, 5). In 48 of 60 studies (80%), the investigated organ-specific AIDs occurred in ≥1% of CSU patients. In contrast, this was the case in only one third of the studies that assessed comorbidity rates of systemic AIDs. The most common organ-specific autoimmune comorbidities in CSU were endocrine (most prevalent: HT), hematological (most prevalent: pernicious anemia) and skin AIDs (most prevalent: vitiligo). Among the systemic AIDs, the most common CSU comorbidites were connective tissue diseases, the most prevalent being RA. In CSU patients, comorbid systemic autoimmune connective tissue diseases were found by 35 studies as compared to only one study describing comorbid autoimmune vasculitis in CSU [12].

3.3. >2% of CSU patients have autoimmune polyglandular syndrome encompassing ATD and vitiligo or pernicious anemia

The rates of comorbidity of two AIDs, ATD and another autoimmune disorder, in CSU patients ranged from 0.9–6.1% (Table 6). Autoimmune polyglandular syndromes (APS) type 3 with ATD and vitiligo (APS type 3C) or ATD and pernicious anemia (APS type 3B) were found in 1.2–5.5% and 5.5–6.1% of CSU patients, respectively (>2% in most studies). The prevalence of overlap syndromes and mixed connective tissue diseases appears to be low and was 0.4–0.5% according to two studies (Table 2).

3.4. Antithyroid and antinuclear antibodies are the most prevalent autoantibodies (AAbs) in CSU

We previously reported that ≥10% of CSU patients exhibit antithyroid AAbs (3.7–37.1% in 24 studies; anti-thyroid peroxidase > anti-thyroglobulin) [9]. In the present analysis, we found that the prevalence of

Table 1
Studies investigating the comorbidity of CSU and AIDs included in the review.

Keywords	Search in Pubmed, n of studies					Additional studies found in the reference lists or Google Scholar	Total number of studies included[b]	Predominant mechanism of the disease
	"Urticaria" keyword	"Chronic urticaria" keyword	Review articles[a]	Case reports[a]	Original papers			
Skin								
Vitiligo	96	37	4	4	7	5	11	AAbs
Psoriasis	444	136	–	1	1	5	5	C
Systemic sclerosis	84	25	–	1	1	2	2	AAbs
Alopecia areata	83	34	–	1	1	–	1	AAbs, C
Pemphigus	91	26	2	1	–	–	0	AAbs
Pemphigoid	90	25	1	–	–	–	0	AAbs
Dermatitis herpetiformis	77	19	–	3	–	–	0	AAbs
Endocrine glands								
Type 1 diabetes mellitus	197	45	1	5	1	5	4	AAbs, C
Autoimmune pancreatitis	3	1	–	–	1	–	1	AAbs
Addison's disease/autoimmune adrenalitis	1/0	1/0	1	–	–	2	0	AAbs
Parathyroiditis	0	0	–	–	–	1	0	AAbs?
Autoimmune hypogonadism	2	0	–	–	–	–	0	AAbs?
Exocrine glands								
Sjögren's syndrome	42	20	1	2	2	7	5	AAbs
Autoimmune oophoritis	0	0	–	–	–	1	0	AAbs
Gastrointestinal system								
Celiac disease/autoimmune enteropathy	58/3	23/2	2	12	8	2	7	AAbs
Primary biliary cirrhosis	6	1	–	4	–	3	3	AAbs
Autoimmune hepatitis	26	16	–	4	1	1	2	AAbs
Autoimmune gastritis	8	6	1	1	1	2	1	AAbs
Primary sclerosing cholangitis	6	–	–	1	–	–	0	AAbs
Crohn's disease	38	9	–	3	–	–	0	C
Ulcerative colitis	52	13	–	3	–	1	0	C
Neurological system								
Lambert-Eaton myasthenic syndrome	2	1	–	1	–	–	0	AAbs
Myasthenia gravis	15	4	–	3	–	–	0	AAbs
Guillain-Barre syndrome	18	4	–	1	–	–	0	C
Multiple sclerosis	41	8	–	1	–	1	0	AAbs, C
Hematological system								
Antiphospholipid syndrome	15	4	–	–	2	–	1	AAbs
Pernicious anemia	8	4	–	2	–	3	0	AAbs
Autoimmune anemia	38	14	–	3	–	–	0	AAbs
Autoimmune thrombocytopenia[d]	30	17	–	–	–	–	0	AAbs
Autoimmune neutropenia	5	1	–	–	–	–	0	AAbs
Kidney								
Glomerulonephritis	129	39	–	5	–	–	0	AAbs
Goodpasture syndrome	0	0	–	–	–	–	0	AAbs
Connective tissue, systemic, and multi-organ								
Rheumatoid arthritis	220	53	–	6	2	10	6	AAbs, C
Systemic lupus erythematosus[c]	376	77	7	4	6	11	6	AAbs
Dermatomyositis	54	18	–	1	1	–	0	AAbs
Palindromic rheumatism	0	0	–	–	–	1	0	AAbs
Sarcoidosis	35	7	–	1	–	–	0	C
Spondyloarthritis	26	5	–	1	–	3	0	AAbs
Rheumatic fever	30	8	–	1	–	0	0	AAbs
Vascular system								
Eosinophilic granulomatosis with polyangiitis	19/1	3/1	–	3	–	6	6	AAbs
Microscopic polyangiitis	2	0	–	1	1	–	1	AAbs
Wegener's granulomatosis	10	2	–	1	–	–	0	AAbs
Polyarteritis nodosa	50	15	–	3	1	–	0	AAbs
Behçet's disease	16	4	–	1	–	1	0	C
Eyes								
Autoimmune uveitis	3	1	–	–	–	–	0	AAbs
Cardiovascular system								
Myocarditis	9	0	–	–	–	–	0	AAbs?
Other								
Autoimmune polyglandular/polyendocrine syndrome	11/0	5/0	1	6	–	2	2	AAbs, C
Total number of studies							64	

AAbs: autoantibody-mediated mechanism and/or the formation of immune complexes; C: cell-mediated.
[a] Excluded or added in the Table 6.
[b] Many studies evaluated the prevalence of several AIDs but were counted only once.
[c] Only studies that reported the prevalence of SLE in CSU were included (the rates of urticarial rash in SLE have been reported earlier [10], Table 5).
[d] Including search with the term "autoimmune thrombocytopenic purpura".

Table 2
The prevalence of autoimmune diseases in CSU.

First author of the study [reference]	Year	Age, years	n, CSU	% (n), disease	Design	Country
Vitiligo						
Kolkhir et al. [25]	2017	≥18	234	3.8 (9)	R	Russia
Akarsu et al. [46]	2015	≥8	146	1.4 (2)	R	Turkey
Vikramkumar et al. [47]	2014	≥18	48	6.2 (3)	CS	India
Lai and Su [12]	2010	≥15	106	0.9 (1)	R	Taiwan
Feibelmann et al. [48]	2007	≥18	49	4.1 (2)	CC	Brazil
Irinyi et al. [49]	2007	≥13	82	9.8 (8)	P	Hungary
Giménez-Arnau et al. [50]	2004	≥18	166	1.2 (2)	P	Spain
Karaayvaz et al. [51]	2002	≥19	60	3.3 (2)	P	Turkey
Leznoff and Sussman [52]	1989	≥8	90	5.5 (5)	–	Canada
Celiac disease						
Magen et al. [53]	2015	≥18	569	4.2 (24)	R	Israel
Confino-Cohen et al. [3]	2012	≥18	12,778	0.5 (64)	R	Israel
Caminiti et al. [18]	2005	≤18	79	5 (4)	CC, CS	Italy
Gabrielli et al. [19]	2005	≥18	80	1.2 (1)	CC	Italy
Caproni et al. [54]	2004	≥19	68	1.5 (1)	R	Italy
Meneghetti et al. [55]	2004	<18	32	9.3 (3)	–	Italy
Liutu et al. [56]	1998	≥13	107	1.9 (2)	D	Finland
Gallo et al. [57]	1992	–	43	7 (3)	–	Italy
Rheumatoid arthritis						
Dionigi et al. [58]	2016	≥18	100	1 (1)	P	Brazil
Akarsu et al. [46]	2015	≥8	146	0.7 (1)	R	Turkey
Magen et al. [53]	2015	≥18	569	0.9 (5)	R	Israel
Confino-Cohen et al. [3]	2012	≥18	12,778	1.5 (187)	R	Israel
Lai and Su [12]	2010	≥15	106	2.8 (3)	R	Taiwan
Irinyi et al. [49]	2007	≥13	82	2.4 (2)	P	Hungary
Caminiti et al. [18]	2005	≤18	95	2.1 (2)	CC, CS	Italy
Harris et al. [59]	1983	<16	86	1.2 (1)	P	UK
Sjögren's syndrome						
Magen et al. [53]	2015	≥18	569	0.5 (3)	R	Israel
Kim et al. [60]	2015	≥18	57	1.7 (1)	CC	Korea
Confino-Cohen et al. [3]	2012	≥18	12,778	0.6 (72)	R	Israel
Lai and Su [12]	2010	≥15	106	4.7 (5)	R	Taiwan
Giménez-Arnau et al. [50]	2004	≥18	166	0.6 (1)	P	Spain
Verneuil et al. [61]	2004	≥18	45[a]	2.2 (1)	P, CC	France
Kozel et al. [62]	2001	≥15	147	0.7 (1)	P	Netherlands
Diabetes mellitus type 1						
Akarsu et al. [46]	2015	≥8	146	5.5 (8)	R	Turkey
Magen et al. [53]	2015	≥18	569	0.2 (1)	R	Israel
Confino-Cohen et al. [3]	2012	≥18	12,778	1.3 (165)	R	Israel
Krupa Shankar et al. [63]	2010	≥10	65	3.1 (2)	P	India
Liutu et al. [56]	1998	≥13	107	1.9 (2)	D	Finland
Autoimmune hepatitis						
Akarsu et al. [46]	2015	≥8	146	0	R	Turkey
Magen et al. [53]	2015	≥18	569	0.2 (1)	R	Israel
Primary biliary cirrhosis						
Pogorelov et al. [64][c]	2016	≥16	230	0.4 (1)	R	Russia
Magen et al. [53]	2015	≥18	569	0.2 (1)	R	Israel
Systemic sclerosis						
Magen et al. [53]	2015	≥18	569	0.3 (2)	R	Israel
Lai and Su [12]	2010	≥15	106	0.9 (1)	R	Taiwan
Asero et al. [65]	2003	≥13	257	0	–	Italy
Pernicious anemia						
Magen et al. [53]	2015	≥18	569	5.4 (31)	R	Israel
Mete et al. [66]	2004	≥18	33	6.1 (2)[b]	CC	Turkey
Leznoff and Sussman [52]	1989	≥8	90	5.5 (5)	–	Canada
Systemic lupus erythematosus						
Akarsu et al. [46]	2015	≥8	146	0.7 (1)	R	Turkey
Magen et al. [53]	2015	≥18	569	0.7 (4)	R	Israel
Colgecen et al. [67]	2015	≥5	237	0.4 (1)	R	Turkey
Kim et al. [60]	2015	≥18	57	1.7 (1)	CC	Korea
Confino-Cohen et al. [3]	2012	≥18	12,778	0.4 (52)	R	Israel
Krupa Shankar et al. [63]	2010	≥10	65	3.1 (2)	P	India
Lai and Su [12]	2010	≥15	106	1.9 (2)	R	Taiwan
Kulthanan et al. [26]	2007	≥15	407	1.2 (5)	R	Thailand
Kozel et al. [62]	2001	≥15	147	0.7 (1)	P	Netherlands
Sibbald et al. [68]	1991	>18	173	0.6 (1)	P	Canada
Champion [69]	1988	>0	1650	0	R	UK
Jacobson et al. [70]	1980	>0	125	0.8 (1)	R	USA

(continued on next page)

Table 2 (*continued*)

First author of the study [reference]	Year	Age, years	n, CSU	% (n), disease	Design	Country
Spondiloarthropathy (ankylosing spondylitis)						
Akarsu et al. [46]	2015	≥8	146	0.7 (1)	R	Turkey
Magen et al. [53]	2015	≥18	569	0.5 (3)	R	Israel
Lai and Su [12]	2010	≥15	106	1.9 (2)	R	Taiwan
Polymyositis/dermatomyositis						
Magen et al. [53]	2015	≥18	569	0	R	Israel
Behcet's disease						
Lai and Su [12]	2010	≥15	106	0.9 (1)	R	Taiwan
Overlap syndromes and mixed connective tissue diseases						
Magen et al. [53]	2015	≥18	569	0.5 (3)	R	Israel
Asero et al. [65]	2003	≥13	257	0.4 (1)	–	Italy
Multiple sclerosis						
Liutu et al. [56]	1998	≥13	107	0.9 (1)	D	Finland
Ulcerative colitis						
Liutu et al. [56]	1998	≥13	107	0.9 (1)	D	Finland
Antiphospholipid syndrome						
Dionigi et al. [58]	2016	≥18	100	1 (1)	P	Brazil
Palindromic rheumatism						
Lai and Su [12]	2010	≥15	106	0.9 (1)	R	Taiwan
Autoimmune adrenalitis (Addison's disease)						
Giménez-Arnau et al. [50]	2004	≥18	166	0.6 (1)	P	Spain
Leznoff and Sussman [52]	1989	≥8	90	2.2 (2)	–	Canada
Parathyroiditis						
Leznoff and Sussman [52]	1989	≥8	90	1.1 (1)	–	Canada
Autoimmune oophoritis						
Giménez-Arnau et al. [50]	2004	≥18	166	0.6 (1)	P	Spain
Psoriasis						
Magen et al. [71]	2013	≥18	1051	3.3 (35)	R	Israel
Silvares et al. [72]	2007	>0	125[a]	0.8 (1)	P, D	Brazil
Liutu et al. [56]	1998	≥13	107	1.9 (2)	D	Finland

R: retrospective study; P: prospective study; CS: cross-sectional study; CC: case-control study; D: descriptive study; —: no data, not defined in the paper or in the abstract.
[a] It is not clear from the paper whether patients with inducible urticaria were excluded.
[b] Patients had low B_{12} levels (not due to diet habits), anti-parietal cell antibodies and atrophic gastritis.
[c] This case report is the part of the retrospective study (results have not been published yet).

antinuclear antibodies (ANA) in CSU patients ranged from 0 to 31.9%, in 38 studies (0–31.9% in adult CSU patients, 21 studies, and 0–19.2% in children with CSU, 6 studies, Table 3). ANA positivity was found in >5% of adult and pediatric CSU patients by 81% (17 of 21 studies) and 33% (2 of 6 studies) of studies, respectively. The rates of CSU patients, who were positive for rheumatoid factor (RF, Table 3), anti-dsDNA, anti-parietal cell antibodies, anti-SSA, anti-SSB, anti-Sm and anti-transglutaminase antibodies, ranged from 0 to 19% (12 studies; ≥1% in 8 of 12 studies), 0–9.9% (7 studies; 0% in 5 studies), 0.2–15.2% (3 studies), 0–7.7% (3 studies), 1.4–14.3% (2 studies), 0–4.4% (2 studies) and 0.2% (1 study), respectively.

3.5. More than 15% of CSU patients have a positive family history for AIDs as reported by most studies

In 10 studies analyzed, the prevalence of a positive family history (FH) for ATD only and for all AIDs with or without ATD ranged from 0 to 19.1% and 0–22.6%, respectively (Table 7). The majority of studies reported a positive FH for AIDs in ≥15% of CSU patients (7 of 10 studies, >12% for ATD in most studies). Pediatric CSU patients reportedly had lower rates of a positive FH of AIDs than adults (0–17.3% vs 15–22.6%). In two studies, none of the pediatric CSU patients had a FH for AIDs [13,14]. Overall, 0–50% (3 studies) and 26.1% (one study) of pediatric and adult CSU patients with autoimmune disorders, respectively, had a FH for AIDs.

3.6. The prevalence of urticarial rash in patients with AIDs is >1% in most studies

In 29 studies, the prevalence of urticarial rash in patients with organ-specific AIDs and systemic AIDs was 0.4–56.7% (12 studies) and 0–56.2% (17 studies), respectively (Tables 4, 5). Three-fourths of these studies (22 of 29) reported urticarial rash including CSU in >1% of AID patients, and more than a half of them in ≥3% of patients (15 of 29 studies). Specifically, this comes from 77% (24 of 31) of all studies on organ-specific AIDs (including 19 studies previously analyzed for comorbidity in patients with ATD [9]) and 78% (46 of 59) of the studies on systemic AIDs (including 42 studies on systemic lupus erythematosus (SLE) previously analyzed [10]).

3.7. Urticarial rash is most prevalent in eosinophilic granulomatosis with polyangiitis, autoimmune thyroid disease, SLE, RA and celiac disease

The rates of urticarial rash were >1.5% in patients with celiac disease or RA, >5% for SLE, >7% for autoimmune thyroid disease (ATD), and ≥10% for eosinophilic granulomatosis with polyangiitis (EGPA, formely known as Churg-Strauss syndrome), in most studies (Tables 4, 5). In two studies, urticarial rash occurred in 9% of patients with APS type 1, which includes chronic candidiasis and/or chronic hypoparathyroidism and/or Addison's disease [15] (Table 4). Urticarial rash, in patients with organ-specific AID, is most prevalent in those with endocrine (ATD) and

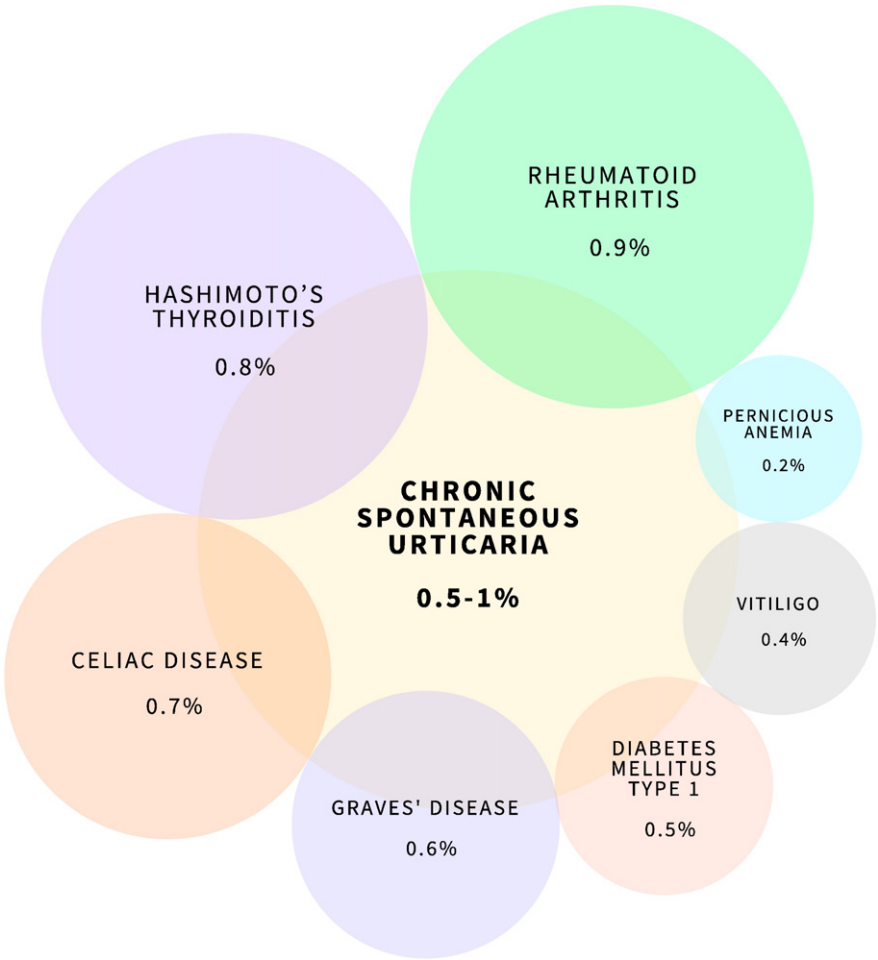

Fig. 1. Venn diagram of the most common autoimmune comorbidity in chronic spontaneous urticaria. The size of the circles roughly represents overall prevalence of autoimmune diseases except CSU and the area of overlap shows reciprocal comorbidity [5,30].

gastrointestinal (celiac disease) AIDs (Table 5). In patients with systemic AIDs, comorbid urticaria is most prevalent in patients with EGPA and SLE.

4. Discussion

4.1. Do CSU patients have an increased risk of autoimmune comorbidity?

The question whether CSU patients are at risk of AIDs, e.g. whether AIDs are more common in CSU patients than individuals without CSU, can not be answered by the studies we analyzed, since most of the studies did not investigate control groups. Hence, we can only address this question by comparing prevalence rates of AIDs in patients with CSU and the general population. This approach, however, has several limitations. For example, many AIDs and CSU are more common in adult females [3,5], and prevalence rates from population studies do not take this into account.

AIDs have an overall estimated prevalence of 4.5% in the general population, with 2.7% for males and 6.4% for females [5]. For individual AIDs, the prevalence is ≤1% [5,16]. In CSU, the prevalence rates of most individual AIDs are ≥1%, i.e. increased (Table 5). ATD, the most frequent organ-specific AID globally (0.62–0.86%) [5], is also the most frequent autoimmune comorbidity in CSU (≥2%).

In a systematic review and a meta-analysis, IgG antithyroid AAbs were found to be higher in CSU patients than in controls (including healthy controls) [9,17]. In a large retrospective study, CSU patients were at significantly higher risk of having ATD, SLE, RA, IDDM, Sjögren's syndrome, and celiac disease compared to the control group [3]. The levels of ANA, RF, anti-TG, anti-parietal cell antibodies, but not anti-dsDNA and anticardiolipin antibodies, were much higher in comparison to controls [3]. In this study, AIDs were diagnosed within 10 years after the diagnosis of CSU in >77% of patients with CSU [3]. Moreover, CSU patients were shown to have a higher risk of having celiac disease than controls in some studies [3,18], but not all [19].

4.2. Do CSU patients with one comorbid AID have an increased risk of developing a second comorbid AIDs?

Patients with one AID have a higher probability of being affected by a second one [20,21]. For example, Somers and co-workers conducted a series of population-based cohort studies and found the risk of a second AID such as RA, ATD and IDDM to be increased in patients who already have one of these AIDs [21]. As of now, the risk for having a second AID in patients with CSU is unknown. In one study, 12.5% of 12,778 CSU patients had 1 comorbid AID, and an additional 2.2% had 2 or more comorbid AIDs [3], which is a higher prevalence than in the general population. For example, in the general population of Italy among 25,885 people, 5.1% had one AID and only 0.2% were affected by two or more AIDs [20].

Table 3
The prevalence of AID-associated antibodies in CSU.

First author of the study [reference]	Year	Age, years	n, CSU	% (n), ANA	% (n), RF	% (n), other AAbs	Design	Country
Mariyath et al. [73]	2017	≥14	100	5 (5)	5 (5)	–	CC	India
Ye et al. [74]	2016	≥19	75	10.7 (8)	–	–	P	Korea
Kim et al. [75]	2016	≥12	38	7.9 (3)	0	–	R	Korea
Magen et al. [76]	2016	≥18	41	12.2 (5)	–	–	P	Israel
Eser et al. [77]	2016	<18	52	19.2 (10)	–	–	P	Turkey
Colgecen et al. [67]	2015	≥5	500[a]	2.4 (12)[a]	–	–	R	Turkey
Magen et al. [53]	2015	≥18	569	16 (91)	16.5 (15/91)[c]	9.9 (9/91) anti-dsDNA, 7.7 (7/91) anti-SSA, 14.3 (13/91) anti-SSB, 4.4 (4/91) anti-Sm[c,e]	R	Israel
Akarsu et al. [46]	2015	≥8	146	6.2 (9)	2.7 (4)	–	R	Turkey
Işık et al. [78]	2014	≤17	56	16.1 (9)	–	–	R	Turkey
Ghaffari et al. [79]	2013	>0	78	0	–	–	D	Iran
Calamita and Calamita [80]	2013	≥18	67	10.5 (7)	–	–	CS	Brazil
Kessel et al. [81]	2012	≥18	45	24.4 (11)	–	–	P, CC	Israel
Viswanathan et al. [82]	2012	≥18	131	29 (38)	–	–	R	USA
Confino-Cohen et al. [3]	2012	≥18	12,778	1.9 (248)	1.8 (233)	0.2 (31) ATG, 0.2 (30) APC, 0.1 (16) anti-dsDNA, 0.1 (14) ACL	R	Israel
Sahiner et al. [13]	2011	≤18	94	13.8 (13)	–	0 anti-dsDNA	R	Turkey
Al-Balbeesi [83]	2011	≥18	68	17.6 (12)	0	–	P, CC	Saudi Arabia
Kilic et al. [84]	2010	≤17	34	0	–	–	P	Turkey
Lai and Su [12]	2010	≥15	106	10.4 (11)	–	7.2 (5/69) anti-SSA, 1.4 (1/69) anti-SSB	R	Taiwan
Krupa Shankar et al. [63]	2010	≥10	65	7.7 (5)	–	–	P	India
Jirapongsananuruk et al. [85]	2010	≤15	94	2.1 (2)	–	–	P	Thailand
Najib et al. [86]	2009	≥16	69	31.9 (22)	–	–	R	USA
Nebiolo et al. [87]	2009	≥16	228	21 (48)	–	–	P	Italy
Kulthanan et al. [26]	2007	≥15	233	15 (34)	–	–	R	Thailand
Buss et al. [88]	2007	≥18	313	3.5 (11)	5 (15/300)	–	R	Germany
Caproni et al. [89]	2005	≥23	28[b]	28.5 (8)	–	7.1 (2) APC, 3.6 (1) API	–	Italy
Sackesen et al. [90]	2004	≤19	17	0	–	0 anti-dsDNA	P	Turkey
Toubi et al. [91]	2004	≥17	139	4 (6)	–	–	P	Israel
Mete et al. [66]	2004	≥18	33	18.2 (6)	–	15.2 (5) APC	CC	Turkey
Verneuil et al. [61]	2004	≥18	45[a]	20 (9)	6.7 (3)	0 anti-dsDNA	P, CC	France
Levy et al. [92]	2003	<18	187	2.7 (5)	–	0 anti-dsDNA[d]	–	Israel
Trachsel et al. [93]	1999	≥8	81[a]	10 (8)[a]	–	–	R	Switzerland
Liutu et al. [56]	1998	≥13	107	0.9 (1)	2.8 (3)	–	D	Finland
Turktas et al. [94]	1997	≥16	94	5.3 (5)	–	0 (anti-dsDNA, anti-SSA, and anti-Sm)	–	Turkey
Sibbald et al. [68]	1991	>18	173	15 (26)[a]	19 (33)[a]	–	P	Canada
Leznoff and Sussman [52]	1989	≥8	76	7.9 (6)	0	–	–	Canada
Lanigan et al. [95]	1984	≥18	25	0	–	0 (AMA, ASMA)	–	UK
Juhlin [96]	1981	>5	330	8 (26)	–	–	–	Sweden
Jacobson et al. [70]	1980	>0	125	0.8 (1)	0	–	R	USA

AMA: antimitochondrial antibodies; ASMA: antismooth muscle antibodies; ATG: anti–transglutaminase IgA antibodies; APC: anti–parietal cell antibodies; anti-dsDNA: anti–double-stranded DNA antibodies; ACL: anticardiolipin antibodies; API: anti-pancreatic islet; ANA: antinuclear antibodies; RF: rheumatoid factor; AAbs: autoantibodies; R: retrospective study; P: prospective study; CS: cross-sectional study; CC: case-control study; D: descriptive study; —: no data, not defined in the paper or in the abstract.

[a] Some patients could have chronic inducible urticaria.
[b] CSU patients with positive autologous serum skin test.
[c] In ANA + CSU patients.
[d] Anti-dsDNA antibodies were negative in three patients who were tested.
[e] Other AAbs: anti-Jo-1 (1.1%), anti-centromere (0%), anti-chromatin (2.2%), anti-ribosomal (1.1%), anti-RNP 68 (3.3%), anti-RNP A (35.2%), anti-Scl-70 (5.5%), anti-SmRNP (5.5%).

Polyautoimmunity can be seen in the setting of autoimmune polyglandular syndromes (APS), multiple autoimmune syndromes (MAS) and overlap syndromes [8]. APS are characterized by autoimmune activity against endocrine and non-endocrine organs [15,22]. A few reports considered CSU as an element of APS type 3C, which also includes ATD together with vitiligo and/or alopecia and/or myasthenia gravis and/or multiple sclerosis [15,22–25]. The fact that CSU patients also exhibit increased rates of comorbid ATD and non-vitiligo AID such as IDDM, pernicious anemia, RA and Sjögren's syndrome (Table 6) may indicate that APS type 3A, 3B and 3D could also include CSU. As of yet, CSU has not been included in the classification of APS 3A, 3B and 3D [15,23]. The prevalence of MAS, i.e. three or more well characterized coexistent AIDs, appears not be increased in CSU patients (Table 6). Confino-Cohen and coworkers reported that ≤0.1% of CSU patients have three or more accompanying AIDs. Overlap syndromes (mixed connective tissue diseases), i.e. the presence of signs and symptoms of two or more diverse connective tissue diseases [5,8], appear to be as rare in CSU as they are in the general population.

4.3. Why is the risk increased in CSU patients to develop comorbid AID?

Women are more likely to develop CSU [3,26,27], and they have a higher prevalence of polyautoimmunity [28]. Female CSU patients have a higher risk of developing comorbid AIDs such as SLE and ATD than men [3,5,9,10], and the prevalence of CSU is higher in female than male ATD patients [29].

The age at onset of AIDs varies widely. For many of them it is typically between 20 and 40 years of age [5], which is the most common age of onset of CSU [30]. ATD and CSU are more common in young adults with CSU and SLE, respectively, than, for example, in children [9,10].

CSU patients have a higher prevalence of a positive family history for all AIDs as well as individual AIDs than the general population (>15% in most studies vs <15% [20,31–33]). For example, the prevalence of a positive family history for ATD in the general population was lower than in patients with CSU (≤6% vs >12% in most studies) [32,33] (Table 7).

Table 4
The prevalence of urticarial rash in autoimmune diseases.

First author of the study [reference]	Year	Age, years	n, disease	% (n), urticarial rash[a]	Design	Country
Eosinophilic granulomatosis with polyangiitis						
Baldini et al. [97]	2009	≥18	38	31.6 (12)	R	Italy
Cohen et al. [98]	2007	≥18	48	12.5 (6)	P	Multicenter
Sable-Fourtassou et al. [99]	2005	≥18	112	18 (20)	CS	Multicenter
Guillevin et al. [100]	1999	≥14	96	8.3 (8)	P, R	France
Reid et al. [101]	1998	≥18	23	26 (6)	R	UK
Lanham et al. [102]	1984	≥18	16	56.2 (9)	R	UK
Literature review [102]	1984	–	66	30.3 (20)	R	UK
Rheumatoid arthritis						
Prakash et al. [103]	2015	≥11	120	1.7 (2)	CS	India
Ergun et al. [104]	2008	≥17	215	0.5 (1)	–	Turkey
Magro and Crowson [105]	2003	≥6	43	7 (3)[b]	R	USA
Yamamoto et al. [106]	1995	≥28	142	4.2 (6)	R	Japan
Vitiligo						
Zhang et al. [44]	2009	>0	5601	0.8 (48)[c]	CS	China
Liu et al. [45]	2005	–	3742	0.5 (17)[c]	R	China
Martis et al. [107]	2002	–	100	1 (1)[c]	P	India
Celiac disease						
Ludvigsson et al. [43]	2013	>0	28,900	1.6 (453)[d]	CC	Sweden
Cataldo et al. [108]	2001	≤15	90	3.3 (3)	CC	Italy
Sjögren's syndrome						
Molina et al. [109]	1985	≥13	45	0[e]	–	USA
Alexander and Provost [110]	1983	≥13	22	4.5 (1)	–	USA
Diabetes mellitus type 1						
Youssef et al. [111]	2016	<18	225	1.8 (4)	CC	Egypt
Systemic sclerosis						
Ghosh et al. [112]	2012	≥18	46	4.3 (2)[c]	CS	India
Alopecia areata						
Kumar et al. [113]	1995	≥20	104	1 (1)	CC	India
Autoimmune hepatitis						
Teufel et al. [114]	2010	≥10	278	0.7 (2)	Cohort	Germany
Primary biliary cirrhosis						
Koulentaki et al. [115]	2006	≥34	49	18.4 (9)	P, CC	Greece
Microscopic polyangiitis						
Kluger et al. [116]	2008	≥18	162	1.2 (2)	R	France
Polyarteritis nodosa						
Kluger et al. [116]	2008	≥18	248	6 (15)	R	France
Psoriasis						
Kumar et al. [117]	2004	<14	419	0.24 (1)	R	India
al-Fouzan and Nanda [118]	1994	<12	190	0.5 (1)	R	Kuwait
Autoimmune pancreatitis						
Hirano et al. [119]	2010	>18	42	2.4 (1)	R	Japan
Antiphospholipid syndrome						
Diógenes et al. [120]	2004	≥12	60	26.7 (16)[f]	P	Brazil
Autoimmune polyglandular syndrome type 1[g]						
Zaidi et al. [121]	2017	>0	23	9 (2)	P	India
Perheentupa [122]	2002	>0	89	9 (8)[h]	–	Finland

R: retrospective study; P: prospective study; CS: cross-sectional study; CC: case-control study; —: no data, not defined in the paper or in the abstract.
[a] Studies do not differentiate clearly between CSU and acute urticaria, chronic inducible urticaria and urticarial vasculitis.
[b] Two cases were characteristic of adult Still's disease.
[c] This study does not differentiate clearly between CSU and chronic inducible urticaria.
[d] 0.3% (79) of patients had chronic urticaria.
[e] 11 (27%) of patients had urticaria vasculitis.
[f] This study does not differentiate clearly between CSU and urticarial vasculitis.
[g] Also called autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED).
[h] Urticarial vasculitis in three patients.

Many AIDs including CSU have been shown to be associated with the major histocompatibility complex (MHC, also called HLA) class I and II alleles [34–38]. For example, HLA DRB1*04 (DR4) alleles were linked to increased susceptibility to RA [38]. Two studies demonstrated an increased frequency of HLA DRB1*04 in chronic urticaria patients as compared to controls [35,39]. As for non-HLA genes, protein tyrosine phosphatase nonreceptor type 22 (PTPN22), a strong susceptibility gene for numerous AIDs, is found to be related to CSU [40]. Furthermore, single nucleotide polymorphisms at position −330 and +166 of IL-2 gene were expressed differentially in CSU in comparison to controls. The haplotype GT of IL-2 at these positions might predispose to the development of AIDs in CSU [41].

Table 5
The summary table on the prevalence of autoimmune diseases in CSU and *vice versa*.

Autoimmune diseases	Prevalence in the general population [5]			Prevalence of autoimmune diseases in CSU			Prevalence of urticarial rash in autoimmune diseases			Predominant mechanism
	Country	%, females	%	n, studies	%, range	%, most studies	n, studies	%, range	%, most studies	
Organ-specific autoimmune diseases										
Hashimoto's thyroiditis[a]	USA	95	0.79	16	0.5–27.5	≥5	19[d]	0.4–56.7[d]	>7[d]	AAbs, C
Pernicious anemia	USA	67	0.15	3	5.4–6.1	≥5	–	–	–	AAbs
Vitiligo	World	52	0.4	9	0.6–9.8	≥3	3	0.5–1	≤1	AAbs
Graves' disease[a]	Denmark	88	0.62	8	0–9.1	≥2	19[d]	0.4–56.7[d]	> 7[d]	AAbs
Celiac disease	USA	57	0.75	8	0.5–9.3	≥1	2	1.6–3.3	>1.5	AAbs
IDDM	World	45	0.48	5	0.2–5.5	≥1	1	1.8	–	AAbs, C
Psoriasis	Denmark	–	0.2	3	0.8–3.3	>1	2	0.2–0.5	≤0.5	C
Addison's disease	World	63	0.01	2	0.6–2.2	≤2	–		–	AAbs
Autoimmune hepatitis	Norway	78	0.02	2	0–0.2	<0.5	1	0.7	–	AAbs
PBC	World	89	0.01	2	0.2–0.4	<0.5	1	18.4	–	AAbs
AH	World	55	<0.0001	1	1.1	–	–		–	AAbs?
Autoimmune oophoritis	World	100	<0.0001	1	0.6	–	–		–	AAbs
Alopecia areata	World	50	0.15	–	–	–	1	1	–	AAbs, C
Autoimmune pancreatitis	World	33	0.0007	–	–	–	1	2.4	–	AAbs
Systemic autoimmune diseases										
EGPA	World	43	0.0002	–	–	–	7	8.3–56.2	≥10	AAbs
SLE	World	88	0.03	12	0–3.1	<1	42[a,b]	0–1.2/0–27.5[a,c]	>1/>5[a,c]	AAbs
RA	USA	75	0.86	8	0.7–2.8	≥1	4	0.5–7	>1.5	AAbs, C
Antiphospholipid syndrome	World	74	0.02	1	1	–	1	26.7	–	AAbs
Sjögren's syndrome	USA	94	0.01	7	0.5–4.7	≤1	2	0–4.5	–	AAbs
Systemic sclerosis	World	92	0.02	3	0–0.9	<1	1	4.3	–	AAbs
SAP	–	–	–	3	0.5–1.9	<1	–	–	–	AAbs
Overlap syndromes/MCTD	Japan	80	0.003	2	0.4–0.5	≤0.5	–	–	–	AAbs
Dermatomyositis	USA	67	0.005	1	0	–	–	–	–	AAbs
Palindromic rheumatism	–	–	–	1	0.9	–	–	–	–	AAbs
Behcet's disease	World	50	0.001	1	0.9	–	–	–	–	C
Multiple sclerosis	World	64	0.06	1	0.9	–	–	–	–	AAbs, C
Ulcerative colitis	World	65	0.03	1	0.9	–	–	–	–	C
Microscopic polyangiitis	World	50	0.002	–	–	–	1	1.2	–	AAbs
Polyarteritis nodosa	World	33	0.006	–	–	–	1	6	–	AAbs
Autoimmune polyglandular syndrome										
APS1	World	57	0.0009	–	–	–	2	9	–	AAbs, C
APS2	USA	75	0.002	–	–	–	–	–	–	AAbs, C
APS3	World	83	<0.0001	7	0.9–6.1	≥2	–	–	–	AAbs, C

ATD: autoimmune thyroid disease; IDDM: insulin-dependent diabetes mellitus; SAP: spondiloarthropathy (ankylosing spondylitis); HT: Hashimoto's thyroiditis; PBC: primary biliary cirrhosis; EGPA: eosinophilic granulomatosis with polyangiitis; RA: rheumatoid arthritis; AIT: autoimmune thyroiditis; SLE: systemic lupus erythematosus; AH: autoimmune hypoparathyroiditis; APS: autoimmune polyglandular syndrome; MCTD: mixed connective tissue disease; AAbs: autoantibody-mediated; C: cell-mediated; —: no data.

[a] Data derived from [9,10].

[b] Studies that included patients with CSU or chronic urticarial rash.

[c] Pediatric/adult patients.

[d] The prevalence of both Hashimoto's thyroiditis and Graves' disease.

To sum up, the increased risk of CSU patients to develop comorbid AID(s) may be due, at least in part, to the female preponderance, the peak age of onset, a positive family history of AIDs, and genetic factors.

4.4. Is the prevalence of CSU increased in patients with AIDs?

The point prevalence of CSU is ≤1% [30], which is lower than the prevalence of urticarial rash in many AIDs (≥1%, Table 5). Lindegard found that the prevalence of urticaria is significantly higher in patients with psoriasis than in the general population of Sweden [42]. A nationwide, population-based study from Korea reported an increased risk of CSU in patients with ATD compared to the control group [29]. In a study of 28,900 patients with biopsy-verified celiac disease, Ludvigsson and co-workers found that patients with celiac disease were at an increased risk of having chronic urticaria before and after the diagnosis of celiac disease [43]. Zhang and others observed that the prevalence of chronic urticaria was significantly elevated in generalized vitiligo patients and their first-degree relatives compared with the population prevalence [44,45]. In summary, there is reasonably strong evidence that the prevalence of urticarial rash is increased in many AIDs.

4.5. Limitations

There is considerable heterogeneity in the design of the studies we analyzed as well as in the patient demographics they included. Many studies were small, some were uncontrolled for referral bias and did not clearly differentiate CSU from acute urticaria, chronic inducible urticaria and/or urticarial vasculitis. Furthermore, in many studies the diagnostic criteria for AIDs were not described. The evaluation of CSU prevalence in AIDs or *vice versa* was not the main aim in many studies. Therefore, the number of patients with CSU or AIDs included in such studies may not be exhaustive. Some AIDs, for example sarcoidosis, ulcerative colitis, Behçet's disease and Crohn's disease, currently discussed as disorders with autoinflammatory features. Our systematic review revealed a high variability in the prevalence of AIDs in CSU between the studies. This imprecision is likely to be due to design-specific issues or a variability in the baseline patient characteristics. The study period (before 2017) is likely to include studies of varying stringency of the diagnostic criteria

Table 6
Polyautoimmunity in CSU.

First author of the study [reference]	Year	Age, years	n, CSU	% (n), AIDs	Polyautoimmunity	Country
Kolkhir et al. [22,25][a]	2017	≥18	234	2.6 (6)	ATD (HT) + vitiligo	Russia
Confino-Cohen et al. [3]	2012	≥18	12,778	2.1 (263)	ATD (HT) + other AID, mostly RA	Israel
Lai and Su [12]	2010	≥15	106	0.9 (1)	ATD + SjS	Taiwan
Feibelmann et al. [48]	2007	≥18	49	4.1 (2)	ATD (HT) + vitiligo	Brazil
Giménez-Arnau et al. [50]	2004	≥18	166	1.2 (2)	ATD (HT) + vitiligo	Spain
Mete et al. [66]	2004	≥18	33	6.1 (2)[b]	ATD + PA[b]	Turkey
Leznoff and Sussman [52]	1989	≥8	90	5.5 (5)	ATD[c] + vitiligo	Canada
Leznoff and Sussman [52]	1989	≥8	90	5.5 (5)	ATD[c] + PA	Canada
Leznoff and Sussman [52]	1989	≥8	90	2.2 (2)	ATD[c] + adrenalitis	Canada
Leznoff and Sussman [52]	1989	≥8	90	1.1 (1)	ATD[c] + parathyroiditis	Canada
Case reports						
Lee et al. [123]	2016	4	1	CR	APS1	USA
De Marchi et al. [23]	2015	51	1	CR	HT + PA + MG + IDDM	Italy
Lundberg et al. [124]	2015	12, 13	2	CR	APS1	USA
Kasznicki and Drzewoski [24]	2014	37	1	CR	HT + IDDM + vitiligo	Poland
Ruggeri et al. [125]	2013	36	1	CR	GD + POF	Italy
Millar and Carson [126]	2012	4	1	CR	APS1	Ireland
Huebner et al. [127]	2012	>12	1	CR	APS1	USA
Ramos-Prol et al. [128]	2011	33	1	CR	APS2	Spain
Hyman et al. [129]	2008	12	1	CR	HT + IDDM	USA
Asero et al. [130]	2005	33	1	CR	GD + IDDM	Italy
Hoffman et al. [131]	2003	3	1	CR	GD + LEMS + alopecia + vitiligo + psoriasis	USA
Dalal et al. [132]	2000	13	1	CR	Psoriasis + juvenile RA	Israel
Levine et al. [133]	1999	11	1	CR	HT + celiac disease	Israel
Midelfart et al. [134]	1983	31	1	CR	HT + vitiligo	Norway
Darley et al. [135]	1977	46	1	CR	MG + SLE	UK

[a] These data are from the retrospective study, which results have not been published yet; CR: case report; APS1: autoimmune polyglandular syndrome type 1; APS2: autoimmune polyendocrine syndrome type 2; PA: pernicious anemia; HT: Hashimoto's thyroiditis; GD: Graves' disease; IDDM: insulin-dependent diabetes mellitus; ATD: autoimmune thyroid disease; SjS: Sjogren's syndrome; RA: rheumatoid arthritis; LEMS: Lambert-Eaton myasthenic syndrome; MG: myasthenia gravis; SLE: systemic lupus erythematosus; POF: premature ovarian failure; AIDs: autoimmune diseases.
[b] Antithyroid antibodies were detected in 6 of 11 patients with CSU and low serum B12 levels (only two patients had low levels of B_{12} and autoimmune gastritis).
[c] Most CSU patients had HT.

for AIDs. Future prospective population-based studies are likely to yield more precise estimates.

5. Unmet needs and conclusions

We identified several needs for further research:

- Do CSU patients affected by a first autoimmune disease have a higher probability of being affected by a second autoimmune disorder? What is the age at the onset of the first and second autoimmune disease? What is the time interval between the development of the 1st and 2nd comorbid autoimmune diseases?
- Can clinical characteristics of CSU, serum or histological markers, be predictors of AIDs development and their activity, duration and prognosis? How many CSU patients with AID-associated autoantibodies develop a clinically significant AID, and how much time does this clinical conversion take?
- What is the pathogenic link between CSU and AIDs? What is the role of AID-associated autoantibodies in the pathogenesis of CSU, and how does their presence affect the course, the clinical characteristics of disease and a response to treatment?
- What are the protective factors preventing the developing of other AIDs in CSU patients?
- In patients with AIDs and CSU, can the treatment of AIDs improve CSU and *vice versa*? Are there emerging treatments that increase

Table 7
The prevalence of a positive family history for autoimmune diseases in CSU patients.

First author of the study [reference]	Year	Age, years	n, CSU	% (n), FH	% (n), AIDs	% (n), FH in CSU patients with AIDs	Design	Country
Mariyath et al. [73]	2017	≥14	100	15 (15)[c]	24 (24)[c]	–	CC	India
Lee et al. [136]	2016	≤16	98	12.2 (12)[c], 16.3 (16)[b]	–	–	P	Singapore
Amin et al. [137]	2015	≥18	186	22.6 (42)[b]	28.3 (34/120)[c], 12 (25/209)[b]	–	R	USA
Sahiner et al. [13]	2011	≤18	100	0[b]	3 (3)[c], 4 (4)[b]	0[b]	R	Turkey
Al-Balbeesi [83]	2011	≥18	68	19.1 (13)[c]	33.8 (23)[c]	26.1 (6)[c]	P, CC	Saudi Arabia
Dodig and Richter [14]	2008	≤18	12	0[b]	8.3 (1)[b]	0[b]	–	Croatia
Garmendia et al. [138]	2004	≥18	32[a]	15.6 (5)[b]	0[a]	–	CC	Venezuela
Brunetti et al. [139]	2004	≤16	52	17.3 (9)[b]	0[b]	–	P	Denmark
Levy et al. [92]	2003	<18	187	–	4.3 (8)[c]	50 (4)[b]	–	Israel
Sabroe et al. [140]	1999	>0	105	19 (20)[b]	–	–	P	UK

R: retrospective study; P: prospective study; CC: case-control study; AIDs: autoimmune diseases; FH: a positive family history for AIDs; —: no data, not defined in the paper or in the abstract.
[a] Patients with AID were excluded or were not seen.
[b] Patients with all AIDs.
[c] Patients with ATD only.

the risk of autoimmune comorbidity? Will they be contraindicated in CSU patients with autoimmune comorbidity?

Answers to these questions will help us to better understand CSU and treat patients. At present, the existing studies indicate that CSU is strongly linked to various AIDs, primarily HT, pernicious anemia, vitiligo, IDDM, GD, celiac disease, and RA. CSU patients have an increased risk of AIDs, especially adult female patients and those with a positive family history and a genetic predisposition for AIDs, who should be screened for signs and symptoms of AIDs.

Take-home messages

- Chronic spontaneous urticaria is strongly linked to various autoimmune diseases, primarily Hashimoto's thyroiditis, pernicious anemia, vitiligo, diabetes mellitus type 1, Graves' disease, celiac disease, and rheumatoid arthritis.
- Patients with chronic spontaneous urticaria are at risk of autoimmune diseases, especially adult female patients and those with a positive family history and a genetic predisposition for autoimmune disorders, who should be screened for signs and symptoms of autoimmune diseases.

Disclosure of potential conflict of interest

The authors declare that they have no relevant conflicts of interest.

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References

[1] Zuberbier T, Aberer W, Asero R, Bindslev-Jensen C, Brzoza Z, Canonica GW, et al. The EAACI/GA(2) LEN/EDF/WAO Guideline for the definition, classification, diagnosis, and management of urticaria: the 2013 revision and update. Allergy 2014; 69(7):868–87.
[2] Kolkhir P, Church MK, Weller K, Metz M, Schmetzer O, Maurer M. Autoimmune chronic spontaneous urticaria: what we know and what we do not know. J Allergy Clin Immunol 2017;139(6):1772–81 e1.
[3] Confino-Cohen R, Chodick G, Shalev V, Leshno M, Kimhi O, Goldberg A. Chronic urticaria and autoimmunity: associations found in a large population study. J Allergy Clin Immunol 2012;129(5):1307–13.
[4] Goldblatt F, O'Neill SG. Clinical aspects of autoimmune rheumatic diseases. Lancet 2013;382(9894):797–808.
[5] Hayter SM, Cook MC. Updated assessment of the prevalence, spectrum and case definition of autoimmune disease. Autoimmun Rev 2012;11(10):754–65.
[6] Fridkis-Hareli M. Immunogenetic mechanisms for the coexistence of organ-specific and systemic autoimmune diseases. J Autoimmune Dis 2008;5:1.
[7] Mastrandrea LD. An overview of organ-specific autoimmune diseases including immunotherapy. Immunol Invest 2015;44(8):803–16.
[8] Anaya JM. The diagnosis and clinical significance of polyautoimmunity. Autoimmun Rev 2014;13(4–5):423–6.
[9] Kolkhir P, Metz M, Altrichter S, Maurer M. Comorbidity of chronic spontaneous urticaria and autoimmune thyroid diseases: a systematic review. Allergy 2017; 72(10):1440–60.
[10] Kolkhir P, Pogorelov D, Olisova O, Maurer M. Comorbidity and pathogenic links of chronic spontaneous urticaria and systemic lupus erythematosus–a systematic review. Clin Exp Allergy 2016;46(2):275–87.
[11] Zhernakova A, Withoff S, Wijmenga C. Clinical implications of shared genetics and pathogenesis in autoimmune diseases. Nat Rev Endocrinol 2013;9(11): 646–59.
[12] Lai K-L, Su C-C. Association of chronic urticaria with rheumatic diseases and thyroid autoimmunity, 21; 2010 277–84.
[13] Sahiner UM, Civelek E, Tuncer A, Yavuz ST, Karabulut E, Sackesen C, et al. Chronic urticaria: etiology and natural course in children. Int Arch Allergy Immunol 2011; 156(2):224–30.
[14] Dodig S, Richter D. Chronic autoimmune urticaria in children. Acta Dermatovenerol Croat 2008;16(2):65–71.
[15] Amerio P, Di Rollo D, Carbone A, Auriemma M, Marra ME, De Remigis P, et al. Polyglandular autoimmune diseases in a dermatological clinical setting: vitiligo-associated autoimmune diseases. Eur J Dermatol 2010;20(3):354–8.
[16] Cardenas-Roldan J, Rojas-Villarraga A, Anaya JM. How do autoimmune diseases cluster in families? A systematic review and meta-analysis. BMC Med 2013;11:73.
[17] Pan XF, JQ Gu, Shan ZY. The prevalence of thyroid autoimmunity in patients with urticaria: a systematic review and meta-analysis. Endocrine 2015;48(3): 804–10.
[18] Caminiti L, Passalacqua G, Magazzu G, Comisi F, Vita D, Barberio G, et al. Chronic urticaria and associated coeliac disease in children: a case-control study. Pediatr Allergy Immunol 2005;16(5):428–32.
[19] Gabrielli M, Candelli M, Cremonini F, Ojetti V, Santarelli L, Nista EC, et al. Idiopathic chronic urticaria and celiac disease. Dig Dis Sci 2005;50(9):1702–4.
[20] Sardu C, Cocco E, Mereu A, Massa R, Cuccu A, Marrosu MG, et al. Population based study of 12 autoimmune diseases in Sardinia, Italy: prevalence and comorbidity. PLoS One 2012;7(3):e32487.
[21] Somers EC, Thomas SL, Smeeth L, Hall AJ. Are individuals with an autoimmune disease at higher risk of a second autoimmune disorder? Am J Epidemiol 2009; 169(6):749–55.
[22] Kolkhir P, Pogorelov D, Kochergin N. Chronic spontaneous urticaria associated with vitiligo and thyroiditis (autoimmune polyglandular syndrome IIIC): case series. Int J Dermatol 2017;56(5):e89–90.
[23] De Marchi SU, Cecchin E, De Marchi S. Autoimmune spontaneous chronic urticaria and generalized myasthenia gravis in a patient with polyglandular autoimmune syndrome type 3. Muscle Nerve 2015;52(3):440–4.
[24] Kasznicki J, Drzewoski J. A case of autoimmune urticaria accompanying autoimmune polyglandular syndrome type III associated with Hashimoto's disease, type 1 diabetes mellitus, and vitiligo. Endokrynol Pol 2014;65(4):320–3.
[25] Kolkhir P, Pogorelov D, Kochergin N. About 3% of chronic spontaneous urticaria patients have vitiligo and thyroiditis (autoimmune polyglandular syndrome IIIC). WAO International Scientific Conference (WISC 2016) Abstracts. World Allergy Organ J 2017;10(1):A11.
[26] Kulthanan K, Jiamton S, Thumpimukvatana N, Pinkaew S. Chronic idiopathic urticaria: prevalence and clinical course. J Dermatol 2007;34(5):294–301.
[27] Nettis E, Pannofino A, D'Aprile C, Ferrannini A, Tursi A. Clinical and aetiological aspects in urticaria and angio-oedema. Br J Dermatol 2003;148(3):501–6.
[28] Rojas-Villarraga A, Amaya-Amaya J, Rodriguez-Rodriguez A, Mantilla RD, Anaya JM. Introducing polyautoimmunity: secondary autoimmune diseases no longer exist. Autoimmune Dis 2012;2012:254319.
[29] Kim YS, Han K, Lee JH, Kim NI, Roh JY, Seo SJ, et al. Increased risk of chronic spontaneous Urticaria in patients with autoimmune thyroid diseases: a nationwide, population-based study. Allergy Asthma Immunol Res 2017;9(4):373–7.
[30] Maurer M, Weller K, Bindslev-Jensen C, Gimenez-Arnau A, Bousquet PJ, Bousquet J, et al. Unmet clinical needs in chronic spontaneous urticaria. A GA(2)LEN task force report. Allergy 2011;66(3):317–30.
[31] Kuo CF, Luo SF, KH Yu, See LC, Zhang W, Doherty M. Familial risk of systemic sclerosis and co-aggregation of autoimmune diseases in affected families. Arthritis Res Ther 2016;18(1):231.
[32] Broadley SA, Deans J, Sawcer SJ, Clayton D, Compston DA. Autoimmune disease in first-degree relatives of patients with multiple sclerosis. A UK survey. Brain 2000; 123(Pt 6):1102–11.
[33] Samuelsson U, Sadauskaite V, Padaiga Z, Ludvigsson J. A fourfold difference in the incidence of type 1 diabetes between Sweden and Lithuania but similar prevalence of autoimmunity. Diabetes Res Clin Pract 2004;66(2):173–81.
[34] Tobon GJ, Pers JO, Canas CA, Rojas-Villarraga A, Youinou P, Anaya JM. Are autoimmune diseases predictable? Autoimmun Rev 2012;11(4):259–66.
[35] O'Donnell BF, O'Neill CM, Francis DM, Niimi N, Barr RM, Barlow RJ, et al. Human leucocyte antigen class II associations in chronic idiopathic urticaria. Br J Dermatol 1999;140(5):853–8.
[36] Calamita Z, Bronhara Pela Calamita A. HLA in patients with chronic spontaneous urticaria who are positive for anti-thyroid antibodies. J Eur Acad Dermatol Venereol 2013;27(5):661–2.
[37] Eyre S, Hinks A, Bowes J, Flynn E, Martin P, Wilson AG, et al. Overlapping genetic susceptibility variants between three autoimmune disorders: rheumatoid arthritis, type 1 diabetes and coeliac disease. Arthritis Res Ther 2010;12(5):R175.
[38] Anaya JM, Correa PA, Mantilla RD, Arcos-Burgos M. Rheumatoid arthritis association in Colombian population is restricted to HLA-DRB1*04 QRRAA alleles. Genes Immun 2002;3(1):56–8.
[39] Öztaş P, Önder M, Gönen S, Oztaş M, Söylemezoğlu O. Is there any relationship between human leucocyte antigen class II and chronic Urticaria? (Chronic Urticaria and HLA Class II). Yonsei Med J 2004;45(3):392–5.
[40] Brzoza Z, Grzeszczak W, Rogala B, Trautsolt W, Moczulski D. PTPN22 polymorphism presumably plays a role in the genetic background of chronic spontaneous autoreactive urticaria. Dermatology 2012;224(4):340–5.
[41] Movahedi M, Tavakol M, Rahmani F, Amirzargar AA, Bidoki AZ, Heidari K, et al. Single nucleotide polymorphisms of IL-2, but not IL-12 and IFN-gamma, are associated with increased susceptibility to chronic spontaneous urticaria. Allergol Immunopathol (Madr) 2017;45(4):333–8.
[42] Lindegard B. Diseases associated with psoriasis in a general population of 159,200 middle-aged, urban, native Swedes. Dermatologica 1986;172(6):298–304.
[43] Ludvigsson JF, Lindelof B, Rashtak S, Rubio-Tapia A, Murray JA. Does urticaria risk increase in patients with celiac disease? A large population-based cohort study. Eur J Dermatol 2013;23(5):681–7.
[44] Zhang Z, SX Xu, Zhang FY, Yin XY, Yang S, Xiao FL, et al. The analysis of genetics and associated autoimmune diseases in Chinese vitiligo patients. Arch Dermatol Res 2009;301(2):167–73.
[45] Liu JB, Li M, Yang S, Gui JP, Wang HY, WH Du, et al. Clinical profiles of vitiligo in China: an analysis of 3742 patients. Clin Exp Dermatol 2005;30(4):327–31.

[46] Akarsu S, İlknur T, Özbağçıvan Ö, Fetil E. Accompanying conditions in patients with chronic spontaneous urticaria and urticarial vasculitis: results of a retrospective study. Türkderm 2015;49(1):18–24.
[47] Vikramkumar AG, Kuruvila S, Ganguly S. Autologous serum skin test as an indicator of chronic autoimmune urticaria in a tertiary care hospital in South India. Indian Dermatol Online J 2014;5(Suppl. 2):S87–91.
[48] Feibelmann TC, Goncalves FT, Daud MS, Jorge Ade S, Mantese SA, Jorge PT. Assessment of association between autoimmune thyroid disease and chronic urticaria. Arq Bras Endocrinol Metabol 2007;51(7):1077–83.
[49] Irinyi B, Szeles G, Gyimesi E, Tumpek J, Heredi E, Dimitrios G, et al. Clinical and laboratory examinations in the subgroups of chronic urticaria. Int Arch Allergy Immunol 2007;144(3):217–25.
[50] Giménez-Arnau A, Ferrer M, Peter H-J, Maurer M, Pujol R. Chronic urticaria: prospective ethiologic study and autoimmune syndrome significance. Actas Dermosifiliogr 2004;95:560–6.
[51] Karaayvaz M, Caliskaner Z, Turan M, Akar A, Ozturk S, Ozanguc N. Levothyroxine versus ketotifen in the treatment of patients with chronic urticaria and thyroid autoimmunity. J Dermatolog Treat 2002;13(4):165–72.
[52] Leznoff A, Sussman GL. Syndrome of idiopathic chronic urticaria and angioedema with thyroid autoimmunity: a study of 90 patients. J Allergy Clin Immunol 1989;84(1):66–71.
[53] Magen E, Waitman DA, Dickstein Y, Davidovich V, Kahan NR. Clinical-laboratory characteristics of ANA-positive chronic idiopathic urticaria. Allergy Asthma Proc 2015;36(2):138–44.
[54] Caproni M, Volpi W, Giomi B, Cardinali C, Antiga E, Melani L, et al. Chronic idiopathic and chronic autoimmune urticaria: clinical and immunopathological features of 68 subjects. Acta Derm Venereol 2004;84(4):288–90.
[55] Meneghetti R, Gerarduzzi T, Barbi E, Ventura A. Chronic urticaria and coeliac disease. Arch Dis Child 2004;89(3):293.
[56] Liutu M, Kalimo K, Uksila J, Kalimo H. Etiologic aspects of chronic urticaria. Int J Dermatol 1998;37(7):515–9.
[57] Gallo C, Vighi G, Schroeder J, Strozzi M, Scibilia J, Colafrancesco M, et al. Chronic urticaria atopic dermatitis and celiac disease. Am J Gastroenterol 1992;87(11):1684.
[58] Dionigi PC, Menezes MC, Forte WC. A prospective ten-year follow-up of patients with chronic urticaria. Allergol Immunopathol (Madr) 2016;44(4):286–91.
[59] Harris A, Twarog FJ, Geha RS. Chronic urticaria in childhood: natural course and etiology. Ann Allergy 1983;51(2 Pt 1):161–5.
[60] Kim JA, Kim S, Kim JE, JY Gu, Yoo HJ, Kang HR, et al. Activation of the intrinsic coagulation pathway in patients with chronic Urticaria. Allergy Asthma Immunol Res 2015;7(5):476–82.
[61] Verneuil L, Leconte C, Ballet JJ, Coffin C, Laroche D, Izard JP, et al. Association between chronic urticaria and thyroid autoimmunity: a prospective study involving 99 patients. Dermatology 2004;208(2):98–103.
[62] Kozel MM, Mekkes JR, Bossuyt PM, Bos JD. Natural course of physical and chronic urticaria and angioedema in 220 patients. J Am Acad Dermatol 2001;45(3):387–91.
[63] Krupa Shankar DS, Ramnane M, Rajouria EA. Etiological approach to chronic urticaria. Indian J Dermatol 2010;55(1):33–8.
[64] Pogorelov D, Olisova O, Nekrasova T, Kolkhir P. Chronic spontaneous urticaria associated with primary biliary cirrhosis: a case report and literature review. J Dtsch Dermatol Ges 2016;14(11):1130–2.
[65] Asero R, Lorini M, Tedeschi A. Association of chronic urticaria with thyroid autoimmunity and Raynaud phenomenon with anticentromere antibodies. J Allergy Clin Immunol 2003;111(5):1129–30.
[66] Mete N, Gulbahar O, Aydin A, Sin AZ, Kokuludag A, Sebik F. Low B12 levels in chronic idiopathic urticaria. J Investig Allergol Clin Immunol 2004;14(4):292–9.
[67] Colgecen E, Ozyurt K, Gul AI, Utas S. Evaluation of etiological factors in patients with chronic urticaria. Acta Dermatovenerol Croat 2015;23(1):36–42.
[68] Sibbald RG, Cheema AS, Lozinski A, Tarlo S. Chronic urticaria. Evaluation of the role of physical, immunologic, and other contributory factors. Int J Dermatol 1991;30(6):381–6.
[69] Champion RH. Urticaria: then and now. Br J Dermatol 1988;119(4):427–36.
[70] Jacobson KW, Branch LB, Nelson HS. Laboratory tests in chronic urticaria. JAMA 1980;243(16):1644–6.
[71] Magen E, Mishal J, Schlesinger M. Clinical and laboratory features of chronic idiopathic urticaria in the elderly. Int J Dermatol 2013;52(11):1387–91.
[72] Silvares MR, Coelho KI, Dalben I, Lastoria JC, Abbade LP. Sociodemographic and clinical characteristics, causal factors and evolution of a group of patients with chronic urticaria-angioedema. Sao Paulo Med J 2007;125(5):281–5.
[73] Mariyath O, Binitha M, Anilakumari V, Biju G, Nikhila P, Pradeep M. Association between chronic spontaneous urticaria and thyroid autoimmunity: a case control study from a tertiary care centre. Int J Res Dermatol 2017;3(1):64–8.
[74] Ye YM, Park JW, Kim SH, Ban GY, Kim JH, Shin YS, et al. Prognostic factors for chronic spontaneous urticaria: a 6-month prospective observational study. Allergy Asthma Immunol Res 2016;8(2):115–23.
[75] Kim DH, Sung NH, Lee AY. Effect of levothyroxine treatment on clinical symptoms in hypothyroid patients with chronic urticaria and thyroid autoimmunity. Ann Dermatol 2016;28(2):199–204.
[76] Magen E, Zueva E, Mishal J, Schlesinger M. The clinical and laboratory characteristics of acute spontaneous urticaria and its progression to chronic spontaneous urticaria. Allergy Asthma Proc 2016;37(5):394–9.
[77] Eser I, Yologlu N, Baydemir C, Aydogan M. The predictive factors for remission of chronic spontaneous urticaria in childhood: outcome from a prospective study. Allergol Immunopathol (Madr) 2016;44(6):537–41.
[78] Işık S, Arıkan Ayyıldız Z, Çağlayan Sözmen Ş, Fırıncı F, Uysal P, Karaman Ö. The etiological evaluation of our patients with chronic urticaria. Türkderm 2014;48(1):13–6.
[79] Ghaffari J, Khademloo M, Mohammadzadeh I, Golpoor M. Chronic Urticaria: the necessity of laboratory examination. Zahedan J Res Med Sci 2013;15(4):66–8.
[80] Calamita Z, Pela Calamita AB. Chronic spontaneous urticaria: epidemiological characteristics focusing on the histocompatibility profile and presence of antibodies. Inflamm Allergy Drug Targets 2013;12(1):8–11.
[81] Kessel A, Yaacoby-Bianu K, Vadasz Z, Peri R, Halazs K, Toubi E. Elevated serum B-cell activating factor in patients with chronic urticaria. Hum Immunol 2012;73(6):620–2.
[82] Viswanathan RK, Biagtan MJ, Mathur SK. The role of autoimmune testing in chronic idiopathic urticaria. Ann Allergy Asthma Immunol 2012;108(5):337–41 [e1].
[83] Al-Balbeesi AO. Significance of antithyroid antibodies and other auto-antibodies in Saudi patients with chronic urticaria. Possible parameters in predicting chronic over three years disease. J Saudi Soc Dermatol Dermatol Surg 2011;15(2):47–51.
[84] Kilic G, Guler N, Suleyman A, Tamay Z. Chronic urticaria and autoimmunity in children. Pediatr Allergy Immunol 2010;21(5):837–42.
[85] Jirapongsananuruk O, Pongpreuksa S, Sangacharoenkit P, Visitsunthorn N, Vichyanond P. Identification of the etiologies of chronic urticaria in children: a prospective study of 94 patients. Pediatr Allergy Immunol 2010;21(3):508–14.
[86] Najib U, Bajwa ZH, Ostro MG, Sheikh J. A retrospective review of clinical presentation, thyroid autoimmunity, laboratory characteristics, and therapies used in patients with chronic idiopathic urticaria. Ann Allergy Asthma Immunol 2009;103(6):496–501.
[87] Nebiolo F, Bergia R, Bommarito L, Bugiani M, Heffler E, Carosso A, et al. Effect of arterial hypertension on chronic urticaria duration. Ann Allergy Asthma Immunol 2009;103(5):407–10.
[88] Buss YA, Garrelfs UC, Sticherling M. Chronic urticaria–which clinical parameters are pathogenetically relevant? A retrospective investigation of 339 patients. J Dtsch Dermatol Ges 2007;5(1):22–9.
[89] Caproni M, Giomi B, Volpi W, Melani L, Schincaglia E, Macchia D, et al. Chronic idiopathic urticaria: infiltrating cells and related cytokines in autologous serum-induced wheals. Clin Immunol 2005;114(3):284–92.
[90] Sackesen C, Sekerel BE, Orhan F, Kocabas CN, Tuncer A, Adalioglu G. The etiology of different forms of urticaria in childhood. Pediatr Dermatol 2004;21(2):102–8.
[91] Toubi E, Kessel A, Avshovich N, Bamberger E, Sabo E, Nusem D, et al. Clinical and laboratory parameters in predicting chronic urticaria duration: a prospective study of 139 patients. Allergy 2004;59(8):869–73.
[92] Levy Y, Segal N, Weintrob N, Danon YL. Chronic urticaria: association with thyroid autoimmunity. Arch Dis Child 2003;88(6):517–9.
[93] Trachsel C, Pichler WJ, Helbling A. Importance of laboratory investigations and trigger factors in chronic urticaria. Schweiz Med Wochenschr 1999;129(36):1271–9.
[94] Turktas I, Gokcora N, Demirsoy S, Cakir N, Onal E. The association of chronic urticaria and angioedema with autoimmune thyroiditis. Int J Dermatol 1997;36(3):187–90.
[95] Lanigan SW, Adams SJ, Gilkes JJ, Robinson TW. Association between urticaria and hypothyroidism. Lancet 1984;1(8392):1476.
[96] Juhlin L. Recurrent urticaria: clinical investigation of 330 patients. Br J Dermatol 1981;104(4):369–81.
[97] Baldini C, Della Rossa A, Grossi S, Catarsi E, Talarico R, d'Ascanio A, et al. Churg-Strauss syndrome: outcome and long-term follow-up of 38 patients from a single Italian centre. Reumatismo 2009;61(2):118–24.
[98] Cohen P, Pagnoux C, Mahr A, Arene JP, Mouthon L, Le Guern V, et al. Churg-Strauss syndrome with poor-prognosis factors: a prospective multicenter trial comparing glucocorticoids and six or twelve cyclophosphamide pulses in forty-eight patients. Arthritis Rheum 2007;57(4):686–93.
[99] Sable-Fourtassou R, Cohen P, Mahr A, Pagnoux C, Mouthon L, Jayne D, et al. Antineutrophil cytoplasmic antibodies and the Churg-Strauss syndrome. Ann Intern Med 2005;143(9):632–8.
[100] Guillevin L, Cohen P, Gayraud M, Lhote F, Jarrousse B, Casassus P. Churg-Strauss syndrome. Clinical study and long-term follow-up of 96 patients. Medicine (Baltimore) 1999;78(1):26–37.
[101] Reid AJ, Harrison BD, Watts RA, Watkin SW, McCann BG, Scott DG. Churg-Strauss syndrome in a district hospital. QJM 1998;91(3):219–29.
[102] Lanham JG, Elkon KB, Pusey CD, Hughes GR. Systemic vasculitis with asthma and eosinophilia: a clinical approach to the Churg-Strauss syndrome. Medicine (Baltimore) 1984;63(2):65–81.
[103] Prakash B, Jayashankar C, Shivalingappa V, Bhakthavatsalam S, Chandrashekar K. Cutaneous manifestations of rheumatoid arthritis. IJRCI 2015;3(1):OA2.
[104] Ergun T, Inanc N, Tuney D, Kotiloglu EK, Seckin D, Tetik C, et al. Skin manifestations of rheumatoid arthritis: a study of 215 Turkish patients. Int J Dermatol 2008;47(9):894–902.
[105] Magro CM, Crowson AN. The spectrum of cutaneous lesions in rheumatoid arthritis: a clinical and pathological study of 43 patients. J Cutan Pathol 2003;30(1):1–10.
[106] Yamamoto T, Ohkubo H, Nishioka K. Skin manifestations associated with rheumatoid arthritis. J Dermatol 1995;22(5):324–9.
[107] Martis J, Bhat R, Nandakishore B, Shetty JN. A clinical study of vitiligo. Indian J Dermatol Venereol Leprol 2002;68(2):92–3.
[108] Cataldo F, Marino V, Di Stefano P. Celiac disease and risk of Atopy in childhood. Pediatr Asthma Allergy Immunol 2001;15(2):77–80.

[109] Molina R, Provost TT, Alexander EL. Two types of inflammatory vascular disease in Sjogren's syndrome. Differential association with seroreactivity to rheumatoid factor and antibodies to Ro (SS-A) and with hypocomplementemia. Arthritis Rheum 1985;28(11):1251–8.
[110] Alexander EL, Provost TT. Cutaneous manifestations of primary Sjogren's syndrome: a reflection of vasculitis and association with anti-Ro(SSA) antibodies. J Invest Dermatol 1983;80(5):386–91.
[111] Youssef RM, Ibrahim A, Amin IM, Soliman HM, Ali A. Cutaneous manifestations among Egyptian children and adolescents with type 1 diabetes. Egypt Pediatr Assoc Gaz 2016;64(1):44–9.
[112] Ghosh SK, Bandyopadhyay D, Saha I, Barua JK. Mucocutaneous and demographic features of systemic sclerosis: a profile of 46 patients from eastern India. Indian J Dermatol 2012;57(3):201–5.
[113] Kumar B, Sharma VK, Sehgal S. Antismooth muscle and antiparietal cell antibodies in Indians with alopecia areata. Int J Dermatol 1995;34(8):542–5.
[114] Teufel A, Weinmann A, Kahaly GJ, Centner C, Piendl A, Worns M, et al. Concurrent autoimmune diseases in patients with autoimmune hepatitis. J Clin Gastroenterol 2010;44(3):208–13.
[115] Koulentaki M, Ioannidou D, Stefanidou M, Maraki S, Drigiannakis I, Dimoulios P, et al. Dermatological manifestations in primary biliary cirrhosis patients: a case control study. Am J Gastroenterol 2006;101(3):541–6.
[116] Kluger N, Pagnoux C, Guillevin L, Frances C. Comparison of cutaneous manifestations in systemic polyarteritis nodosa and microscopic polyangiitis. Br J Dermatol 2008;159(3):615–20.
[117] Kumar B, Jain R, Sandhu K, Kaur I, Handa S. Epidemiology of childhood psoriasis: a study of 419 patients from northern India. Int J Dermatol 2004;43(9): 654–8.
[118] al Fouzan AS, Nanda A. A survey of childhood psoriasis in Kuwait. Pediatr Dermatol 1994;11(2):116–9.
[119] Hirano K, Tada M, Isayama H, Kawakubo K, Yagioka H, Sasaki T, et al. Clinical analysis of high serum IgE in autoimmune pancreatitis. World J Gastroenterol 2010; 16(41):5241–6.
[120] Diogenes MJ, Diogenes PC, de Morais Carneiro RM, Neto CC, Duarte FB, Holanda RR. Cutaneous manifestations associated with antiphospholipid antibodies. Int J Dermatol 2004;43(9):632–7.
[121] Zaidi G, Bhatia V, Sahoo S, Sarangi AN, Bharti N, Zhang L, et al. Autoimmune polyendocrine syndrome type 1 in an Indian cohort: a longitudinal study. Endocr Connect 2017;6(5):289–96.
[122] Perheentupa J. APS-I/APECED: the clinical disease and therapy. Endocrinol Metab Clin North Am 2002;31(2):295–320 [vi].
[123] Lee J, Rodrigues J, Knutsen A, Myers S, Becker B. P335 Autoimmune polyendocrinopathy candidiasis ectodermal dystrophy presenting with chronic autoimmune urticaria and onychomycosis. Ann Allergy Asthma Immunol 2016; 117(5):S121.
[124] Lundberg C, Martinez B, Banks TA. Expanding the spectrum: chronic urticaria and autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy. Ann Allergy Asthma Immunol 2015;114(4):353–4.
[125] Ruggeri RM, Vita G, D'Angelo AG, Quattrocchi P, Certo R, Benvenga S, et al. The unusual association of Graves' disease, chronic spontaneous urticaria, and premature ovarian failure: report of a case and HLA haplotype characterization. Arq Bras Endocrinol Metabol 2013;57(9):748–52.
[126] Millar S, Carson D. Clinical phenotypes of autoimmune polyendocrinopathycandidiasis-ectodermal dystrophy seen in the Northern Ireland paediatric population over the last 30 years. Ulster Med J 2012;81(3): 118–22.
[127] Huebner S, Nelson M, Martinez B. Chronic Urticaria as possible manifestation of Autoimmune Polyendocrinopathy-Candidiasis-Ectodermal Dystrophy (APECED). J Allergy Clin Immunol 2012;129(2):AB216.
[128] Ramos-Prol A, Rubio-Almanza M, Campos-Alborg V, Febrer-Bosch I, Merino-Torres JF. Chronic autoimmune urticaria as a possible non endocrine manifestation of autoimmune polyglandular syndrome type II. Endocrinol Nutr 2011; 58(9):503–5.
[129] Hyman SJ, Shreffler WG, Rapaport R. Type 1 diabetes, autoimmune thyroid disease, and chronic urticaria. Pediatr Diabetes 2008;9(5):508–11.
[130] Asero R, Orsatti A, Tedeschi A, Lorini M. Autoimmune chronic urticaria associated with type 1 diabetes and Graves' disease. J Allergy Clin Immunol 2005;115(5): 1088–9.
[131] Hoffman WH, Helman SW, Sekul E, Carroll JE, Vega RA. Lambert-Eaton Myasthenic syndrome in a child with an autoimmune phenotype. Am J Med Genet A 2003; 119A(1):77–80.
[132] Dalal I, Levine A, Somekh E, Mizrahi A, Hanukoglu A. Chronic urticaria in children: expanding the “autoimmune kaleidoscope”. Pediatrics 2000;106(5): 1139–41.
[133] Levine A, Dalal I, Bujanover Y. Celiac disease associated with familial chronic urticaria and thyroid autoimmunity in a child. Pediatrics 1999; 104(2):e25.
[134] Midelfart K, Moseng D, Kavli G, Stenvold SE, Volden G. A case of chronic urticaria and vitiligo, associated with thyroiditis, treated with PUVA. Dermatologica 1983; 167(1):39–41.
[135] Darley CR, Munro DD, Borrie PF. Myasthenia gravis, systemic lupus erythematosus and urticaria. (Case 1). Myasthenia gravis and herpetiform acantholysis. (Case 2). Proc R Soc Med 1977;70(10):734–7.
[136] Lee XH, Ong LX, Cheong JY, Sultana R, Rao R, Lim HH, et al. A stepwise approach in the management of chronic spontaneous urticaria in children. Asia Pac Allergy 2016;6(1):16–28.
[137] Amin P, Levin L, Holmes SJ, Picard J, Bernstein JA. Investigation of patient-specific characteristics associated with treatment outcomes for chronic urticaria. J Allergy Clin Immunol Pract 2015;3(3):400–7.
[138] Garmendia JV, Zabaleta M, Blanca I, Bianco NE, De Sanctis JB. Total and biologically active serum-soluble CD154 in patients with chronic idiopathic urticaria. Allergy Asthma Proc 2004;25(2):121–5.
[139] Brunetti L, Francavilla R, Miniello VL, Platzer MH, Rizzi D, Lospalluti ML, et al. High prevalence of autoimmune urticaria in children with chronic urticaria. J Allergy Clin Immunol 2004;114(4):922–7.
[140] Sabroe RA, Seed PT, Francis DM, Barr RM, Black AK, Greaves MW. Chronic idiopathic urticaria: comparison of the clinical features of patients with and without anti-FcepsilonRI or anti-IgE autoantibodies. J Am Acad Dermatol 1999;40(3):443–50.